

AI Pneumonia Detector - Project Review Guide

1. Project Overview

The **AI Pneumonia Detector** is a medical web application that analyzes uploaded chest X-rays to detect probabilities of Pneumonia. This project was designed with three core goals:

1. **Accessibility:** It must be a seamless web interface accessible from anywhere.
 2. **Efficiency:** It must run entirely on free-tier cloud hosting with strict memory barriers (Max 512MB RAM).
 3. **Medical Safety:** The system must intelligently block users from uploading selfies, documents, or screenshots by verifying the biological anatomy of the image before passing it to the AI.
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2. Technologies Used & Why

Frontend: React.js (Vite)

- **Why we used it:** React allows us to break the UI down into reusable parts (like the `UploadBox`). Vite is a modern build tool that makes React incredibly fast to load.
- **How it helps us:** Instead of statically refreshing the page every time the user clicks "Predict," React silently fetches the data from the backend and instantly renders the result on the screen.

Backend: FastAPI (Python)

- **Why we used it:** FastAPI is one of the fastest Python frameworks. It is asynchronous, meaning it doesn't freeze the server when waiting for an image to download.
- **How it helps us:** It effortlessly manages the endpoints that accept the heavy image files (`UploadFile`), validates them, and bridges the gap to our Machine Learning model.

AI Model format: TensorFlow Lite (TFLite)

- **Why we used it:** Standard AI models (like `.h5` Keras models) are massive. Loading a standard AI model requires hundreds of megabytes of RAM. Since our Render cloud server only gives us 512MB RAM, the system would aggressively crash (OOM limit) every time we ran a prediction.
- **How it helps us:** We mathematically converted our AI model to **TensorFlow Lite**. TFLite statically sets up tiny memory buffers, drastically reducing our active server memory from ~400MB to barely ~50MB. *It mathematically does the exact same prediction but with zero bloat!*

Deployment: Vercel & Render

- **Vercel:** Hosts the pristine React frontend perfectly globally.
 - **Render:** Runs our entire Python/TFLite backend server on the cloud.
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3. Strict Image Gatekeeping (Computer Vision)

One of the most complex modules in this project is our **Image Validation layer**. If a user uploads a photo of a dog or a black-and-white screenshot, the AI would blindly try to "diagnose" it. We fixed this by mathematically verifying an image is actually an X-Ray using pure Python Computer Vision before the AI ever sees it!

We do this in three strict checks:

A. The Color Heuristics Test

- **What it does:** Real X-Rays are purely grayscale or faintly tinted. Standard photos have thousands of varying deep colors.

- **How it works:** We calculate the Standard Deviation (`np.std`) of all the RGB color channels spanning the image.
- **Simple Example:** Imagine a screenshot of a blue button. The blue button is 100% Blue and 0% Red. Our math detects that massive color gap, flags the standard deviation as too high, and immediately blocks the screenshot.

B. The Anatomical Layout Test

- **What it does:** All front-facing Chest X-rays have the same biological shape: A bright dense object directly down the middle (your Spine & Heart), surrounded by dark air-filled bags on the left and right (your Lungs).
- **How it works:** We slice the image array into 3 vertical columns (Left, Center, Right). We average the brightness of the clusters.
- **Simple Example:** Imagine someone uploads a black-and-white selfie of them standing outdoors against a bright white sky. The math calculates that the Center (the dark person) is darker than the sides (the bright sky). This violates biological X-ray anatomy—so the server rejects it!

C. The High-Frequency Edge Test

- **What it does:** Real X-Rays are soft, blurry, and overlapping body tissue. Photography naturally contains intensely sharp boundaries—like individual hairs, grass, sharp text, or UI rectangles.
- **How it works:** We rapidly run a discrete derivative matrix (`np.diff()`) across the image. This counts the ratio of "sharp transitions".
- **Simple Example:** If you upload a black-and-white PDF document, the math counts thousands of incredibly sharp boundaries around every single letter. The edge density spikes to 25%, proving it's not a blurry soft-tissue X-ray (which naturally rests around <10%). It rejects the document!

Conclusion

By bridging **FastAPI with a restricted TFLite hardware model**, we completely solved server crashes. By integrating **Structural Computer Vision Mathematics**, we ensured total input integrity against spoiled images or natural photography.